

Lab 3: The Phases of the Moon

The Moon is Earth's constant companion and a familiar sight to all of us. While we may take passing notice of the Moon's phases, few of us stop to consider how the geometry and motions of the Moon, Earth, and Sun conspire to create this celestial pattern. Many people are surprised to notice the Moon in the sky during the daytime! In this lab, you'll investigate how the observed phase of the Moon depends on the orientations of Earth and Sun.

Learning Objectives

At the completion of this lab, you should be able to:

- describe the geometry and scale of the Earth-Moon-Sun system
- estimate time of day/night for an observer given the position and rotation of Earth relative to the Sun.
- describe the orbits and rotations of Earth and the Moon, including timescales.
- infer the approximate time of day/night given the Moon's phase and position in the sky.
- sketch the view of the Moon from Earth given its position relative to the Sun and Earth.
- explain why lunar and solar eclipses do not occur during every lunar cycle.

Earth's Motion

On the last page of this lab is the beginning of a drawing of the Earth-Moon-Sun system as viewed from above Earth's northern hemisphere. In addition to answering the questions, you will add details to your drawing as you go along.

Label Earth and the Moon and draw lines to indicate the orbit of each body (Earth's orbit doesn't need to be a complete circle, but the Moon's does).

At what position/longitude on Earth is it local noon in your sketch? Describe it below and draw a stick figure at that location.

Which direction is east and which direction is west from where your stick figure is standing? (Remember, we're looking down on the North Pole.) Draw a "W" and an "E" for those directions.

The Sun appears to rise in the east and set in the west. Does Earth rotate from west to east or east to west? Is this clockwise or counterclockwise in your drawing? Draw an arrow to mark the rotation.

How long does it take for Earth to make one rotation relative to the Sun, in hours? What time is it for an observer on the opposite side of Earth from your stick figure? Label both times in your sketch.

What time is it for observers halfway between the labeled times? (Hint: is the Sun rising or setting for a person in those locations?) Label those times as well.

Earth orbits the Sun in the same direction that it rotates. Indicate the direction of Earth's orbital motion with an arrow. Does Earth orbit clockwise or counterclockwise in your sketch?

Moon's Motion

The Moon orbits and rotates in the same direction as Earth. Indicate these motions with arrows on your diagram.

In fact, the Moon's orbital and rotational periods are equal. On your sketch, place an X on the side of the Moon facing Earth. Add three more Moons to your sketch, showing the Moon's location when it is one quarter, one half, and three quarters of the way through its orbit. How far has the Moon rotated in each position? Indicate where the X is located at each orbital position.

Roughly how long does it take for the Moon to orbit Earth? How many times does Earth rotate in the same period?

Illumination of the Moon

Notice the arrows representing incoming sunlight on your diagram. Shade the parts of each Moon that are not illuminated by the Sun.

How much of the Moon's total surface is illuminated at each position?

Notice whether the X is illuminated at each position of the Moon. Is there a "Dark Side of the Moon?" Why or why not?

But wait - why doesn't Earth block the Sun's light from reaching the Moon when they're aligned Sun-Earth-Moon?

Let's pause a minute to consider the true sizes and distances in the Earth-Moon-Sun system.

Earth's diameter is 12,742 km and the Moon's is 3,474 km. About how many times larger is Earth than the Moon in diameter? The average distance between Earth and the Moon is 384,399 km. About how many times the Earth's diameter is this distance? Sketch the Earth-Moon system to scale below.

The average distance between Earth and the Sun is 149,598,261 km. How many times the average Earth-Moon distance is this?

How far away would the Sun be in your scale drawing of the Earth-Moon system above?

Based on your answers above, why can we consider the Sun's rays to be parallel lines?

The Moon's orbit around Earth is tilted by 5° relative to Earth's orbit around the Sun. Sketch an edge-on view of the system below and explain why the Moon is illuminated even when it is on the other side of Earth from the Sun.

Phases of the Moon

How much of the Moon's surface is potentially visible (either illuminated or not) from Earth at any given time? Explain.

Draw a line through the Moon at each position to separate the part that is visible to Earth from the part that is out of Earth's view.

Sketch the Moon's appearance to an observer on Earth at each position. If you need help visualizing this, use a small ball to represent the Moon and a light source for the Sun-you are Earth.

How much of the illuminated portion of the Moon can be seen from Earth during the full moon phase? New moon?

What fraction of the illuminated portion of the Moon can be seen during the "quarter" phases? Which position is first quarter and which is last quarter? (Remember which direction the Moon orbits the Earth!)

Label each phase with its name in your sketch.

The Moon's positions between the four main phases also have names. The names are combinations of *waxing* or *waning* for when the Moon is "growing or "shrinking" and *crescent* or *gibbous* for when the visible part of the Moon is less than half illuminated or more than half illuminated, respectively (e.g., "waxing crescent" or "waning gibbous").

Label the positions of these phases on your diagrams using arrows between the main phases.

Consider the orbital period of the Moon relative to the rotation period of Earth. Approximately how long does each of the eight phases last for an observer on Earth?

Does everyone on Earth see the same phase of the Moon on a given day/night? Explain.

A lunar eclipse occurs when the Moon passes through Earth's shadow. During what phase can lunar eclipses occur?

A solar eclipse occurs when the Moon blocks the Sun as viewed from the Earth. During what phase can solar eclipses occur?

Why don't lunar and solar eclipses occur during every cycle of phases?

Can you see the Moon during the day?

Combine the time labels on Earth, the phase sketches, and the rotation of Earth to figure out what time of day the Moon rises and sets at each position and what time it is highest overhead. Fill in the table below.

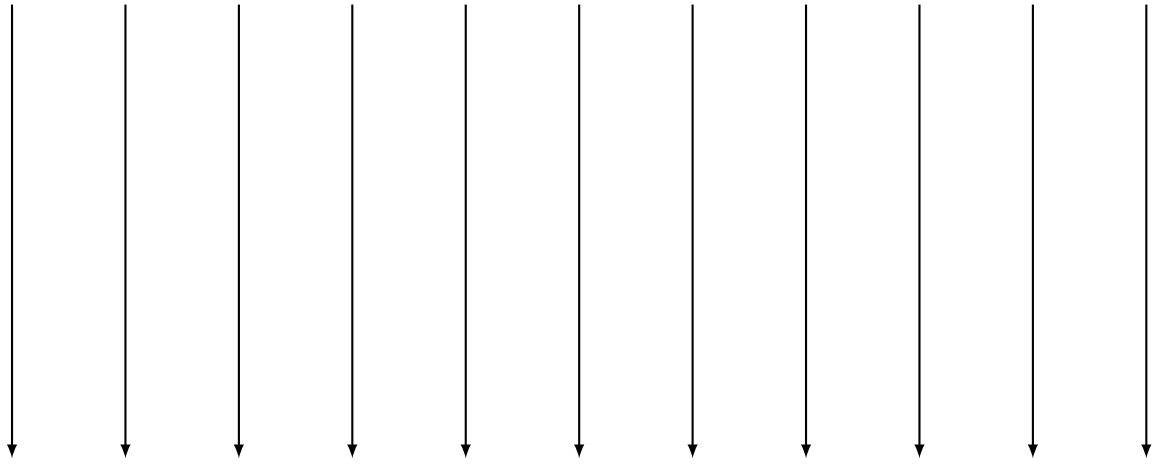
Phase	Rises	Highest Overhead	Sets

Can you see the Moon during the day? What can you say about the Moon's phase if it is visible during the day?

What phase will only be high overhead at midnight?

If you see the Moon directly above a tree from your bedroom window one day, will the Moon appear in the same position earlier or later the next day? Explain.

Phases of the Moon



Incoming Sunlight

